Program : **Diploma in Chemical Engineering**	
Course Code : **6072B**	Course Title: **Project Management - Process Plants**
Semester : **6**	Credits: **4**
Course Category: **Open Elective**	
Periods per week: **4 (L: 3 T: 1 P: 0)**	Periods per semester: **60**

Course Objectives:

- To enable the students to understand the concepts of project management from planning to execution of projects.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Industrial Mgt& Safety		Industrial Mgt& Safety	5

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the responsibilities of a project engineer for engineering and other technical activities relating to the project	12	Understanding
CO2	Summarize the feasibility analysis in project Management and network analysis tools for cost and time estimation	18	Understanding
CO3	Summarize the engineering design and selection for various process equipments	18	Understanding
CO4	Summarize the hazards and safety management in process plants	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1						3	
CO2						3	
CO3						3	
CO4						3	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	**Summarize the responsibilities of a project engineer for engineering and other technical activities relating to the project**		
M1.01	Summarize the scope and role of project engineer	2	Understanding
M1.02	Summarize Technical and Economical Feasibility Report	3	Understanding
M1.03	Summarize the factors involved in construction projects	3	Understanding
M1.04	Illustrate the various process flow diagrams	4	Understanding
Contents: Scope of project engineering, the role of project engineer, pilot plants, commercial, semi commercial plants. R&D, TEFR-contents and format of presenting the feasibility report. Plant location and site selection- factors to be considered, site data, site clearance, preliminary data for construction projects, process engineering, cost estimates, process design procedure, process engineer vs project engineer. Flow diagrams- classification, schematic diagrams, plant layout or plot plans, engineering design and drafting - types of process plant drawings, drafting practices.			
CO2	**Summarize the feasibility analysis in project Management and network analysis tools for cost and time estimation**		
M2.01	Summarize the various tools and techniques for project management	4	Understanding
M2.02	Summarize the project management techniques- CPM and PERT	5	Understanding
M2.03	Outline the various procurement operations	5	Understanding
M2.04	Summarize the project contracts	4	Understanding
	Series Test – I		

Contents: Planning and scheduling of projects, tools and techniques for project management, bar charts - types -horizontal, grouped, stacked. Network techniques-PERT, CPM. Framework for PERT/CPM networks, Activity, event, Drawing the CPM network, Tabulation and analysis of activities, PERT calculations, Applications, Crashing project. Procurement operations- Inquiry, Quotation, bid comparison, purchase order, inspection, expediting- office procedures. Contracts and contractors, contract classification- Cost plus with fixed fee contract, Lump-sum or fixed price contract, Guaranteed Maximum contract- project financing, statutory sanctions.			
CO3	**Summarize the engineering design and selection of various process equipments**		
M3.01	Summarize the engineering design and equipment selection for pressure vessels	4	Understanding
M3.02	Summarize the engineering design and equipment selection for heat exchangers	6	Understanding
M3.03	Summarize the engineering design and equipment selection for process pumps	4	Understanding
M3.04	Summarize the engineering design and equipment selection for motors and turbines	4	Understanding
Contents: Details of engineering design and equipment selection of pressure vessels- types, material selection criteria, design procedure, ASME pressure vessel code, design calculations excluded. Details of engineering design and equipment selection for heat exchangers. Details of engineering design and equipment selection of process pumps-compressors and vacuum pumps. Details of engineering design and equipment selection of motors and turbines, other process equipments.			
CO4	**Summarize the hazards and safety management in process plants**		
M4.01	Summarize the piping fundamentals	3	Understanding
M4.02	Summarize the various hazards in process plants	3	Understanding
M4.03	Summarize the safety in Process design	2	Understanding
M4.04	Summarize the construction planning and commissioning activities	4	Understanding
	Series Test – II		

Contents:
Piping fundamentals- terminology in piping- piping design- pipe size- piping systems- Schedule number- Material of construction- Tests for piping. Hazards in process plant - Physical hazards- Chemical hazards- Mechanical hazards- Electrical hazards- Buildings- Partially enclosed structure- Unenclosed structure. Safety in Process design – inherent safety- passive safety systems- active safety systems- procedural safety - Design in overall safety Management. Construction planning- Construction operations – Commissioning - Mechanical completion, Precommissioning - Precommissioning activities - Control system commissioning - Typical sequence of process commissioning - Start up and commissioning - post commissioning documentation.

Text / Reference

T/R	Book Title/Author
T1	Project Engineering of Process Plants; M.Rase and J.Barrow
R1	Process Equipment Design; Y. Brounell and C. Young
R2	F. Plummer, Project Engineering: Elsevier, 2007.
R3	Perry's Chemical Engineers Hand book; Perry R.H. and. Green D.W

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/110/104/110104073/